

Thesis Progress Update

Thesis Topic: Benchmarking Holistic vs Sentence-Level RLAIIF for Factual Summarization via Granular Credit Assignment (GCA)

Date: 05 May 2026 | Weekly focus: reward models, GCA architecture, and DPO preference construction

1. Questions from Last Meeting

Question / Concern	Current Answer
Can an LLM judge be unfair or inconsistent?	Yes. The main implementation will not depend on an open-ended LLM judge as the primary oracle. A fixed, testable factuality or reward model will be used for preference construction. LLM judging can remain only as a secondary diagnostic.
How is a 4-sentence summary compared with a 6-sentence summary?	The decision is not based on sentence count. Each summary is compared by source-supported information and factual errors. A longer summary can win if it adds correct information; it loses if it adds hallucinations or contradictions.
What if both summaries contain the same facts but different wording?	No factual preference is created. GCA should return a tie or no-preference decision, and the pair is not used for DPO training.
Why do sentence-level segmentation if a full-summary factuality score can be obtained at once?	Full-summary scoring is useful as the holistic baseline, but it gives only one score and does not show where the factual problem is. Sentence-level or claim-level segmentation localizes supported, contradicted, and unsupported information. GCA then aggregates these local scores into a transparent summary-level preference.
Where does GCA come in?	GCA comes after local scoring and before DPO. It aggregates sentence-level or claim-level faithfulness scores into one summary-level chosen/rejected decision.
What does DPO do after GCA?	DPO trains the summarization model using the chosen/rejected pair produced by GCA. It does not directly use the local scores; it learns to assign higher probability to the chosen summary for the same article.

2. Implementation Position

The accepted proposal remains unchanged: the thesis compares holistic feedback with granular feedback aggregated through GCA for factual summarization. The implementation refinement is limited to the judge component. Instead of relying mainly on open-ended LLM judgments, the primary automatic feedback source will be a fixed factuality or reward model.

This keeps the original RLAIIF/GCA setup intact because the preference labels are still produced automatically by an AI model and converted into DPO-compatible preference pairs.

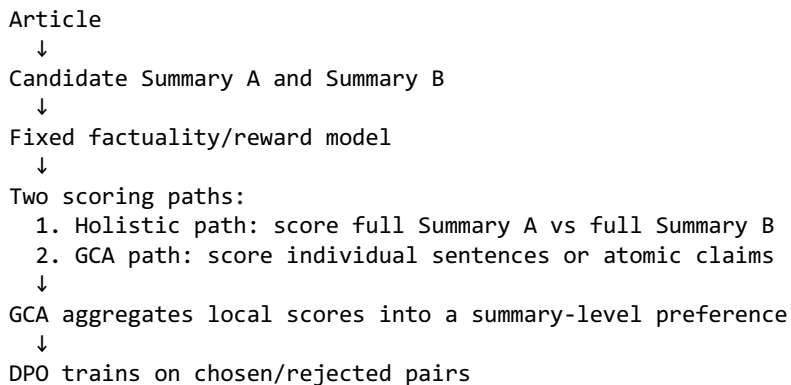
3. Reward and Factuality Model Strategy

The preferred setup is to use one fixed model for preference construction, then validate the resulting models with independent factuality metrics and a small human audit. This avoids building the thesis around a single reward model.

Model / Resource	Role in Thesis	Reason for Use
AlignScore	Strong candidate for primary local scorer	Designed for context-claim factual consistency; fits sentence or claim-level GCA well.
CogComp BART faithful summary detector	Summarization-specific baseline or primary scorer if easier to run	Scores whether a summary is faithful to the article; fixed and reproducible.
RewardBench	Model-selection reference, not the main summarization scorer	Supports the use of benchmarked reward models instead of relying on subjective judging.
SummaC / QAFactEval / FActScore-style checks	Independent evaluation and validation	Reduces circularity by evaluating with metrics different from the training-label source.

4. Updated Architecture

The architecture keeps the same experimental comparison but makes the judge more reproducible.



5. Why Sentence-Level Segmentation is Needed

A full-summary factuality score is useful, but it only gives a final number. It does not explain which sentence or claim caused the lower score. Since factual errors in summaries are often local, segmentation makes the preference signal easier to inspect and easier to filter.

In this thesis, full-summary scoring remains the holistic baseline. The proposed GCA path uses sentence-level or claim-level scores so that credit and blame can be assigned to specific parts of the summary before creating the DPO pair.

6. GCA Scoring Rule

For each summary pair, GCA compares aggregated faithfulness scores. A margin is used so that very small differences do not create artificial preferences.

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GCA_A = average faithfulness score of Summary A units
GCA_B = average faithfulness score of Summary B units
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If GCA_A > GCA_B + margin: choose A
If GCA_B > GCA_A + margin: choose B
Otherwise: no preference
```

7. Short Examples

Example A: 4 sentences vs 6 sentences

If Summary A has four faithful sentences and Summary B has six sentences but includes claims such as “approval already happened” when the article says “approval is still needed,” Summary A is preferred. The reason is factual consistency, not length.

```
A scores: 0.96, 0.94, 0.91, 0.90 → GCA_A = 0.9275
B scores: 0.94, 0.93, 0.91, 0.90, 0.18, 0.35 → GCA_B = 0.7017
Decision: Summary A is chosen; Summary B is rejected.
```

Example B: same facts, different wording

If both summaries express the same source-supported facts in different wording, GCA should produce no preference.

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GCA_A = 0.94
GCA_B = 0.94
Decision: no preference; the pair is discarded from DPO training.
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8. DPO Role

DPO starts only after GCA has produced a clear chosen/rejected pair. It trains the policy model to assign higher probability to the chosen summary than to the rejected summary for the same article. If GCA produces no preference, the pair is skipped.

9. Work Plan for This Week

- Run a small test set with full-summary scoring and sentence-level scoring.
- Compare AlignScore and the CogComp BART faithful summary detector for feasibility and stability.
- Implement the first GCA rule using mean score plus a margin threshold.
- Create a small set of qualitative examples where holistic scoring and GCA agree, disagree, or return no preference.
- Select the primary scorer and keep independent metrics for validation.

Resources

- Lambert et al. RewardBench: Evaluating Reward Models for Language Modeling. arXiv:2403.13787.
- CogComp/bart-faithful-summary-detector. Hugging Face model card.
- Zha et al. AlignScore: Evaluating Factual Consistency with a Unified Alignment Function. ACL 2023.
- Min et al. FActScore: Fine-grained Atomic Evaluation of Factual Precision in Long Form Text Generation. EMNLP 2023.
- Laban et al. SummaC: Re-Visiting NLI-based Models for Inconsistency Detection in Summarization. TACL 2022.
- Fabbri et al. QAFactEval: Improved QA-Based Factual Consistency Evaluation for Summarization. NAACL 2022.